

Evidence of execution

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos DBA

01_star_schema_dimensions.sql

ARCHITECTURE NOTE (DAMA-DMBOK Ch.5 - Data Modeling, Ch.6 - Data Storage): These dimensions are NOT built on top of the raw fragmented source tables (ventas.CLIENTE, facturacion.CLIENTE, flota.CONDUCTOR/EMPLEADO/OPERADOR).

They are built on top of the GOVERNED master data produced in folder 05-reference-and-master-data (governance_control.MASTER_CLIENTE, governance_control.MASTER_CONDUCTOR). This is intentional: a star schema is a consumption layer, not a cleansing layer. If dimensions consumed the raw duplicated sources, every KPI built on top would inherit the same "how many unique clients do we have" problem that justified this entire project (see 02-base-system-scripts and 03-data-quality-diagnostic).

All dimensional tables are physically placed on the ANALYTICS filegroup, isolating dimensional/analytical I/O from OLTP I/O on PRIMARY.

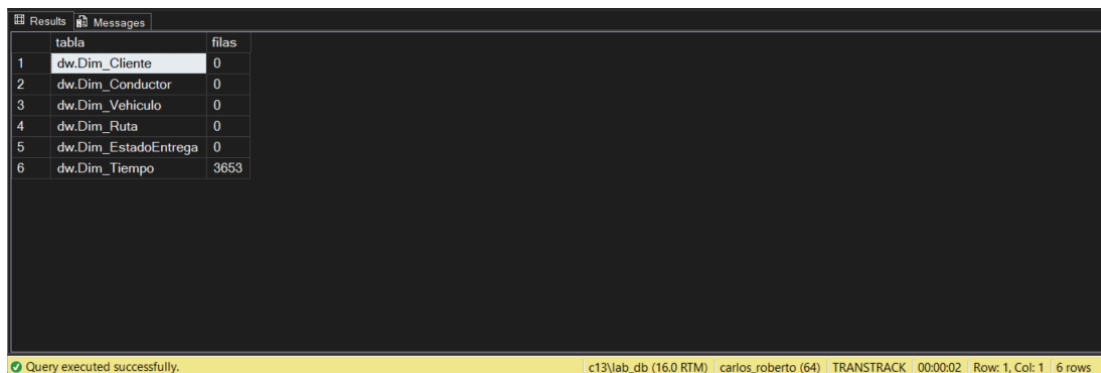


	tabla	filas
1	dw.Dim_Cliente	0
2	dw.Dim_Conductor	0
3	dw.Dim_Vehiculo	0
4	dw.Dim_Ruta	0
5	dw.Dim_EstadoEntrega	0
6	dw.Dim_Tiempo	3653

Query executed successfully. c13\lab_db (16.0 RTM) carlos_roberto (64) TRANSTRACK 00:00:02 Row: 1, Col: 1 6 rows

02_star_schema_facts.sql

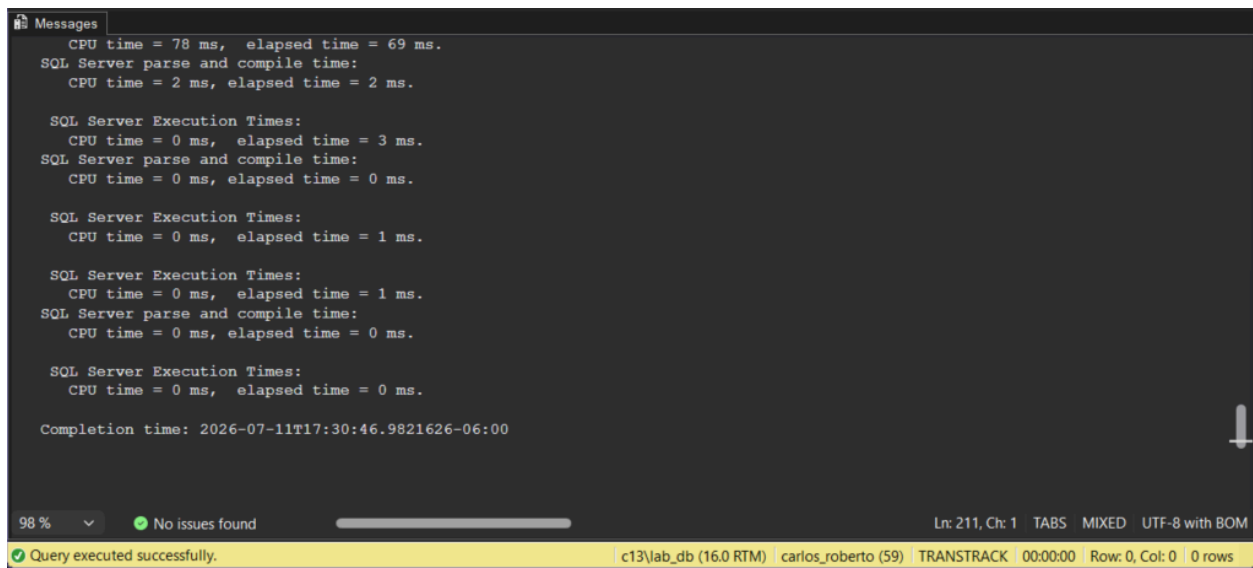
GRAIN DECLARATIONS (mandatory Kimball discipline - state the grain before writing a single column):

- Fact_Pedido: one row per PEDIDO (order placed).
- Fact_Entrega: one row per ENTREGA (physical delivery attempt/result).

- Fact_Incidente: one row per INCIDENTE (accident/theft/delay/damage event).

ARCHITECTURE NOTE - the "point-in-time" join problem:

operaciones.ENTREGA and operaciones.INCIDENTE do not store conductor_id or vehiculo_id directly. That relationship lives in flota.ASIGNACION_VEHICULO, which is itself a dated range (fecha_inicio/fecha_fin) tied to pedido_id. Loading Fact_Entrega/Fact_Incidente therefore requires resolving "which driver/vehicle was assigned to this order AT THE TIME the event happened" - a classic point-in-time dimensional join, not a simple FK lookup. This is documented explicitly in 03_scd2_load_procedure.sql. Getting this join wrong is exactly the kind of silent data-quality defect this whole project exists to prevent (see 03-data-quality-diagnostic).



```
Messages
CPU time = 78 ms, elapsed time = 69 ms.
SQL Server parse and compile time:
  CPU time = 2 ms, elapsed time = 2 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 3 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 1 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 1 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-07-11T17:30:46.9821626-06:00

98 %  No issues found  Lrn: 211, Ch: 1  TABS  MIXED  UTF-8 with BOM
Query executed successfully.  c:\3\lab_db (16.0 RTM)  carlos_roberto (59)  TRANSTRACK  00:00:00  Row: 0, Col: 0  0 rows
```

03_scd2_load_procedure.sql

Pattern: hash-diff comparison + MERGE. Rather than comparing every Type-2 column individually (which forces a table scan because no index can be built on "any of these 5 columns changed"), we precompute a single HASHBYTES('MD5', ...) of the concatenated Type-2 attributes at read time and store it as hash_diff. Detecting a change becomes an equality check on one fixed-length column, which the source query can seek on and which keeps the comparison logic in one place instead of N places.

DBA PRACTICE NOTE: run this procedure with SET STATISTICS XML ON once after the initial load to confirm the plan uses the nonclustered index IX_Dim_Cliente_Actual (WHERE es_version_actual = 1) for the "find current row" step, rather than scanning the whole SCD2 history table as it grows.

```
Messages
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 14 ms.

SQL Server Execution Times:
  CPU time = 15 ms, elapsed time = 11 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 9 ms.

SQL Server Execution Times:
  CPU time = 16 ms, elapsed time = 2 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 2 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 2 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 4 ms.

98 %  No issues found
Ln: 55, Ch: 1 TABS MIXED UTF-8 with BOM
Query executed successfully. c:\lab_db (16.0 RTM) carlos_roberto (59) TRANSTRACK 00:00:09 Row: 0, Col: 0 0 rows
```

04_data_vault_hubs.sql

WHY A SEPARATE MODEL FROM THE STAR SCHEMA (see `adr_datavault_vs_kimball.md` for the full ADR): the star schema answers "how many/how much/trend" business questions and is optimized for BI tools. It is NOT designed to answer "produce every fact known about incident #4521, as it existed at each point in time, admissible as evidence" - that is a fundamentally different requirement (auditability, non-destructive history, hash-based business-key independence) that Data Vault is purpose-built for.

Hash keys (not natural keys, not identity integers) are used per Data Vault 2.0 standard so that:

- 1) Hubs can be loaded in parallel from multiple sources without an identity-column bottleneck or lookup round-trip.
- 2) The same business key computes to the same hash regardless of which source system it arrived from - critical here because CONDUCTOR is fragmented across 3 legacy tables with 3 different natural keys (`numero_licencia`, `codigo_empleado`, `id_operador`).

```
Messages
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 7 ms.
Table 'sysclsobjs'. Scan count 0, logical reads 2, physical reads 1, page server reads 0, read-ahead reads 0, page server read-

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 1 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 7 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 4 ms.
SQL Server parse and compile time:
  CPU time = 78 ms, elapsed time = 91 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

98 %  No issues found  Ln: 270, Ch: 1  TABS  MIXED  UTF-8 with BOM
Query executed successfully.  c13\lab_db (16.0 RTM)  carlos_roberto (59)  TRANSTRACK  00:00:00  Row: 0, Col: 0  0 rows
```

05_data_vault_links.sql

LINK_INCIDENTECONDUCTOR

Resolves which driver was linked to a given incident. Sourced the same way as the point-in-time join in 03_scd2_load_procedure.sql: through flota.ASIGNACION_VEHICULO at the incident's fecha_incidente.

```
Messages
  CPU time = 0 ms, elapsed time = 8 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 2 ms.
SQL Server parse and compile time:
  CPU time = 2 ms, elapsed time = 2 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 1 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 2 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 3 ms.
SQL Server parse and compile time:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

Completion time: 2026-07-11T17:46:26.3694279-06:00

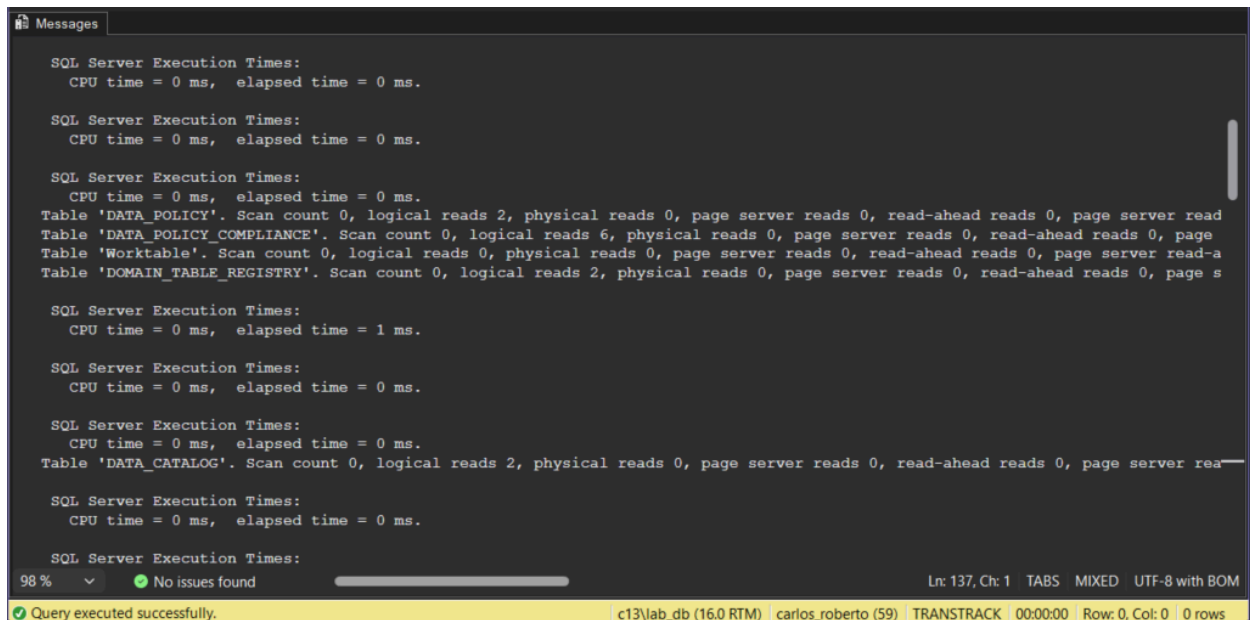
98 %  No issues found  Ln: 184, Ch: 1  TABS  MIXED  UTF-8 with BOM
Query executed successfully.  c13\lab_db (16.0 RTM)  carlos_roberto (68)  TRANSTRACK  00:00:00  Row: 0, Col: 0  0 rows
```

06_data_vault_satellites.sql

SAT_CONDUCTOR_DETALLE

License and employment history for a driver. This is what makes it possible to answer, months or years after the fact, "was this driver's license valid on the date of the incident" - a question the OLTP tables

(which only store current state) cannot answer once the license record has been updated.



```
SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.
Table 'DATA_POLICY'. Scan count 0, logical reads 2, physical reads 0, page server reads 0, read-ahead reads 0, page server read
Table 'DATA_POLICY_COMPLIANCE'. Scan count 0, logical reads 6, physical reads 0, page server reads 0, read-ahead reads 0, page
Table 'Worktable'. Scan count 0, logical reads 0, physical reads 0, page server reads 0, read-ahead reads 0, page server read-a
Table 'DOMAIN_TABLE_REGISTRY'. Scan count 0, logical reads 2, physical reads 0, page server reads 0, read-ahead reads 0, page s

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 1 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.
Table 'DATA_CATALOG'. Scan count 0, logical reads 2, physical reads 0, page server reads 0, read-ahead reads 0, page server rea

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.

SQL Server Execution Times:
  CPU time = 0 ms, elapsed time = 0 ms.
```

98 % No issues found Ln: 137, Ch: 1 TABS MIXED UTF-8 with BOM

✓ Query executed successfully. c13\lab_db (16.0 RTM) carlos_roberto (59) TRANSTRACK 00:00:00 Row: 0, Col: 0 0 rows

07_run_data_loads.sql

RUN THIS AFTER: 01, 02, 03, 04, 05, 06 have already been executed (those files only CREATE tables/procedures - this file is what actually POPULATES them).

resultado		filas_afectadas
1	dw.Dim_Cliente SCD2 load completed.	0

tabla	filas
1 dw.Dim_Cliente	12984
2 dw.Dim_Conductor	1200
3 dw.Dim_Vehiculo	800
4 dw.Dim_Ruta	2388
5 dw.Dim_EstadoEntrega	5

tabla	filas
1 dw.Fact_Pedido	432644
2 dw.Fact_Entrega	415329

tabla	filas
1 vault.Hub_Conductor	1200
2 vault.Hub_Vehiculo	800
3 vault.Hub_Pedido	500000
4 vault.Hub_Incidente	8473

tabla	filas
1 vault.Link_IncidenteConductor	0
2 vault.Link_IncidenteVehiculo	1256
3 vault.Link_IncidentePedido	7411
4 vault.Sat_Conductor_Detalle	0
5 vault.Sat_Incidente_Detalle	0

Query completed with errors. c13\lab_db (16.0 RTM) carlos_roberto (59) TRANSTRACK 00:03:39 Row: 1, Col: 1 17 rows

08_incident_reconstruction.sql

his is the query that justifies building a Data Vault at all. It answers a question the star schema is not designed to answer well: "walk me through everything connected to incident #X, as it was known at that time, with a defensible trail back to source."

USAGE: replace @IncidenteId below with the incidente_id under investigation.

seccion	incidente_id	fecha_incidente	tipo_incidente	severidad	estado_resolucion	costo_estimado	descripcion	
seccion	master_conductor_id	nombre_completo	numero_licencia	categoria_licencia	fecha_vencimiento_lic	licencia_vigente_en_ese_momento	version	
seccion	vehiculo_id	placa	marca	modelo	tipo_vehiculo	capacidad_ton		
seccion	pedido_id	numero_pedido	tipo_carga	peso_kg	valor_declarado	master_cliente_id	cliente_nombre	nit_venta
1	PEDIDO_Y_CLIENTE	359643	PED-000358238	MENSAJERIA	21407.00	11506.00	3449	Comercial Pacifico Ltda. 6303 NIT-006
seccion	fecha_hora	latitud	longitud	velocidad_kmh	rumbo	evento		

Query executed successfully.

c13\lab_db (16.0 RTM)

carlos_roberto (62)

TRANSTRACK

00:01:43

Row: 1, Col: 3

1 rows